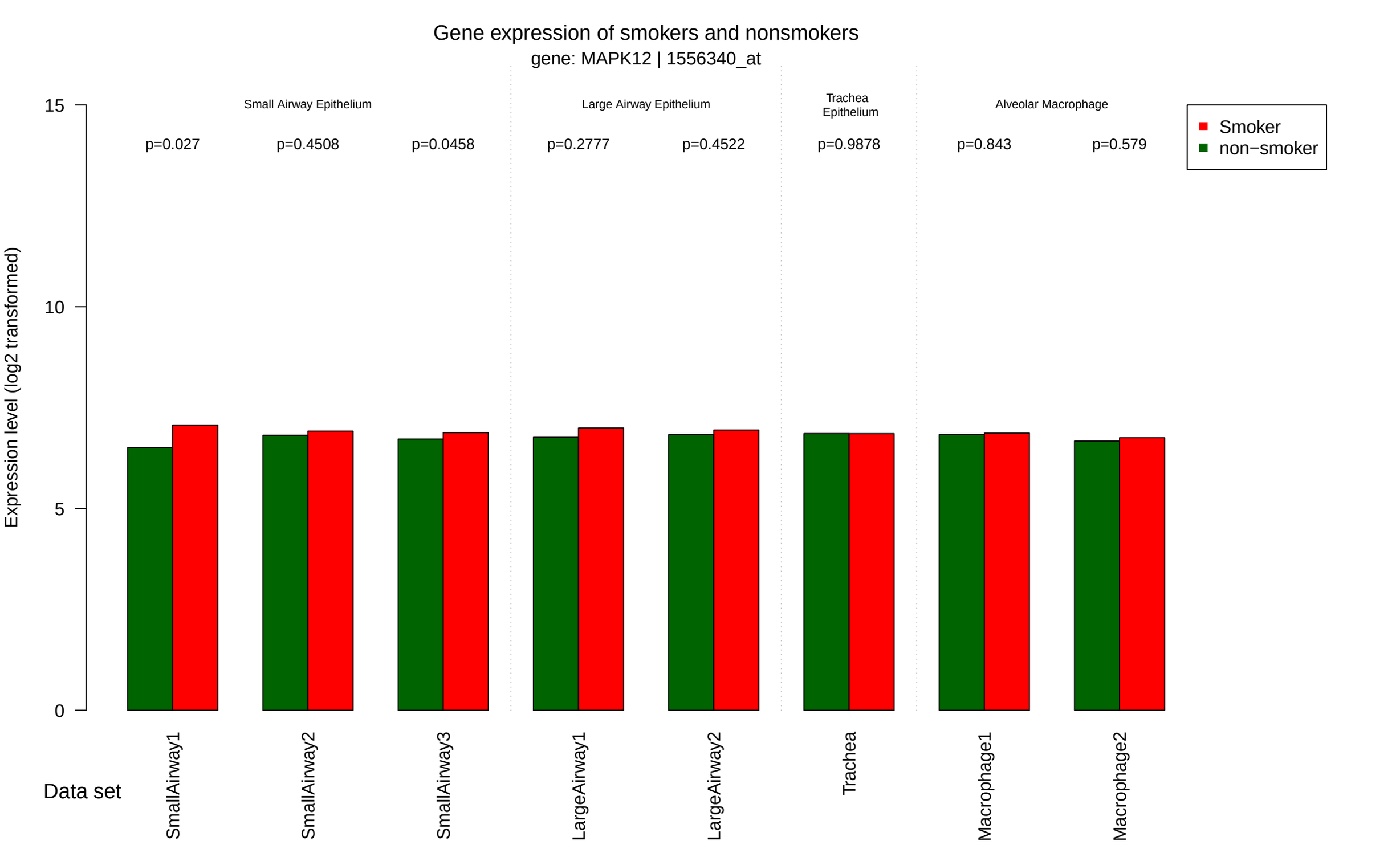
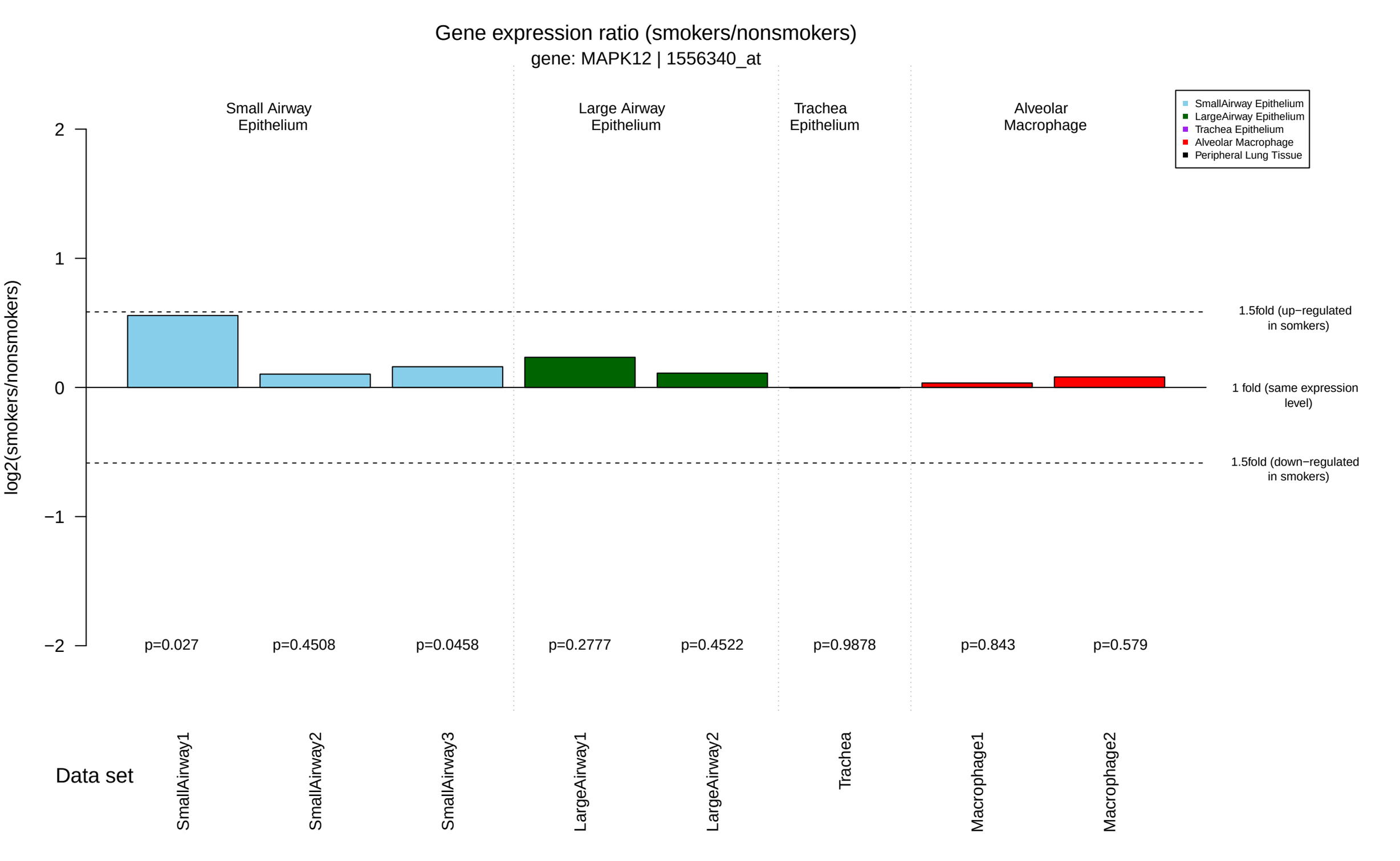
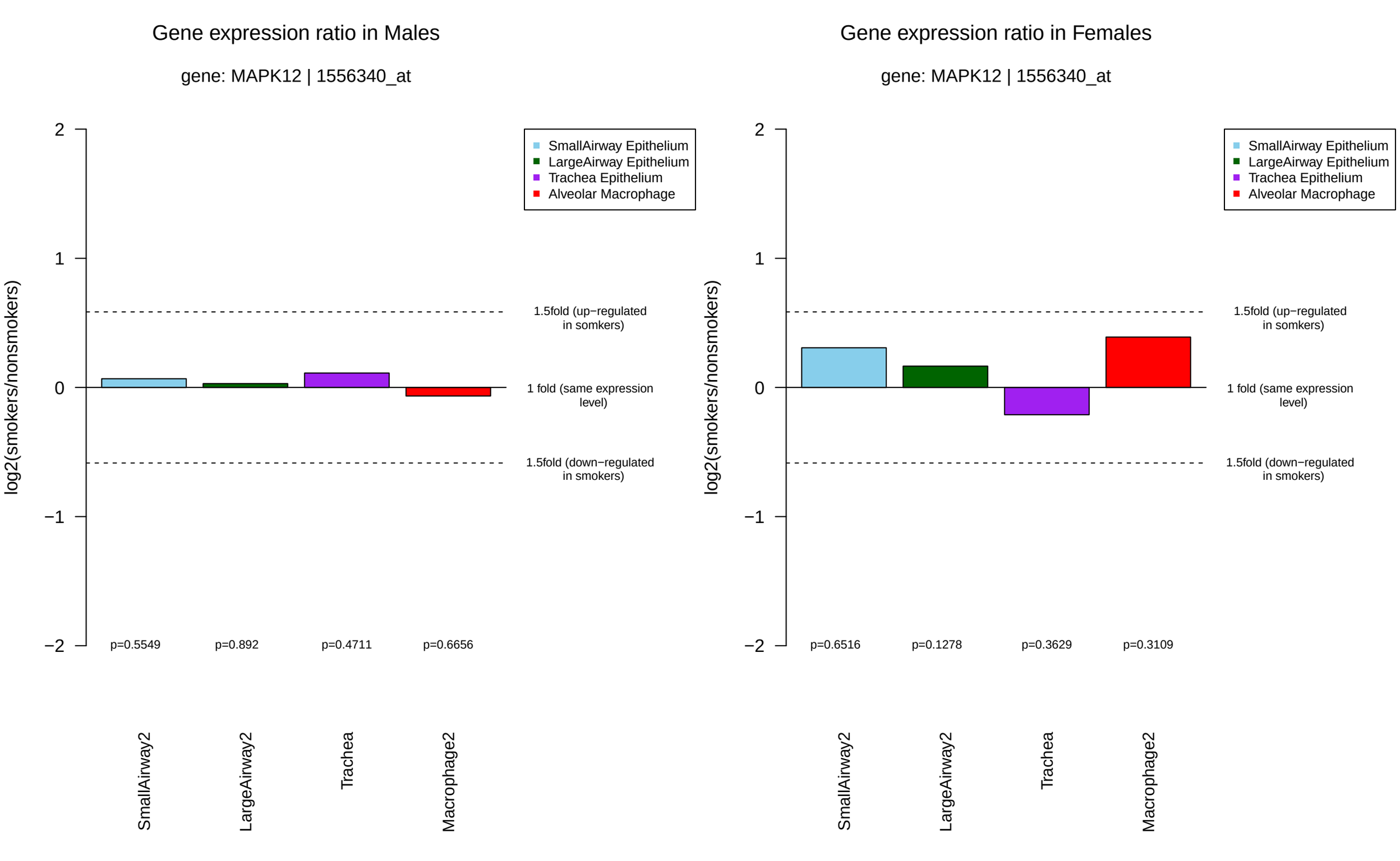
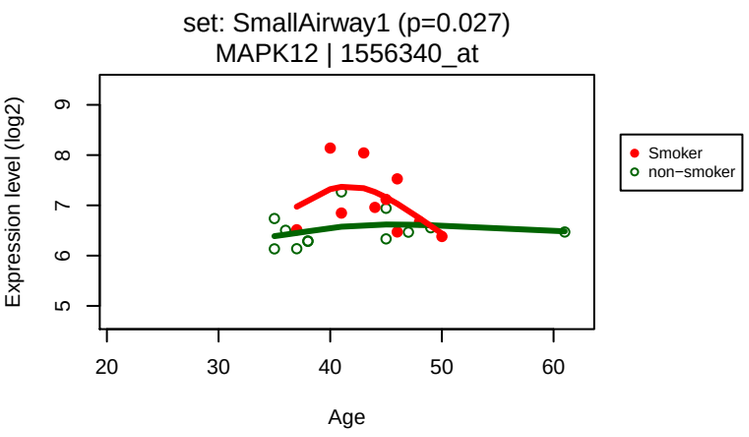
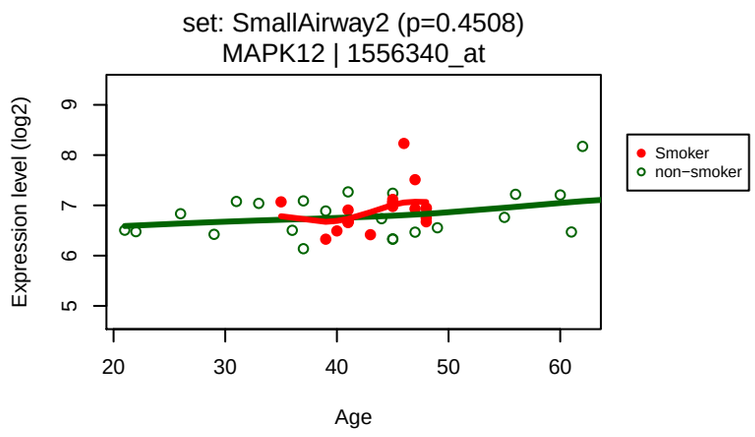
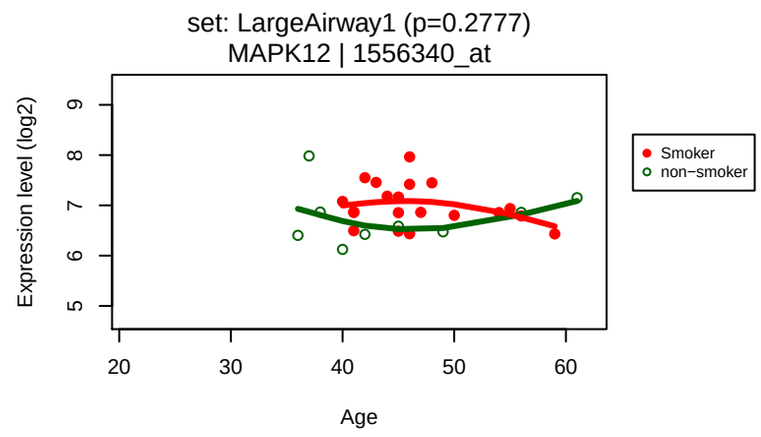

Gene expression of smokers and nonsmokers
gene: MAPK12 | 1556340_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.027
p=0.4508
p=0.0458
p=0.2777
p=0.4522
p=0.9878
p=0.843
p=0.579
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

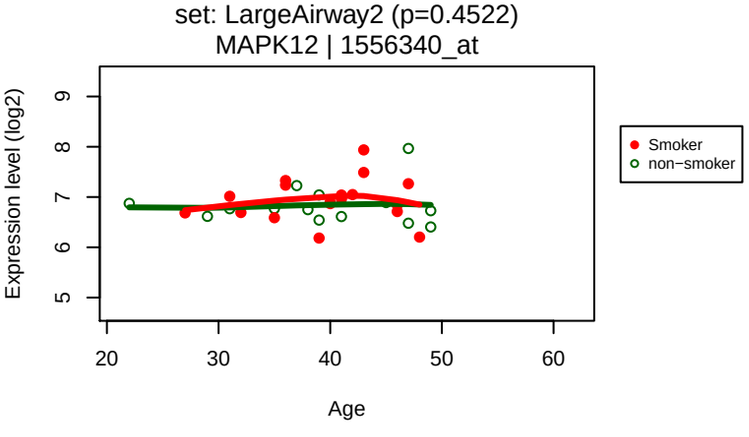

Gene expression ratio (smokers/nonsmokers)
gene: MAPK12 | 1556340_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.027
p=0.4508
p=0.0458
p=0.2777
p=0.4522
p=0.9878
p=0.843
p=0.579
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

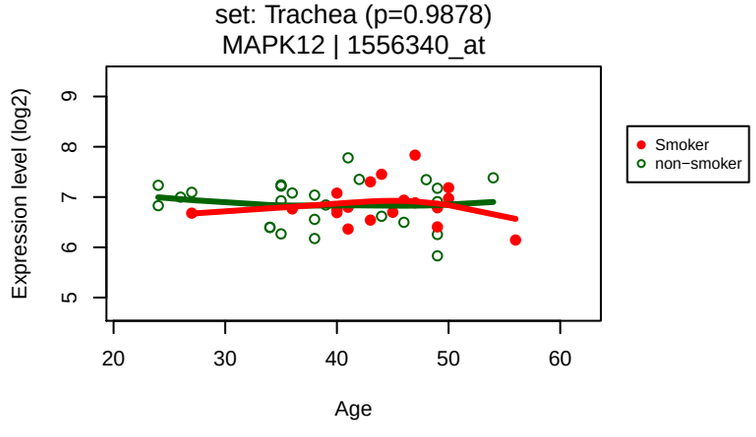

Gene expression ratio in Males
gene: MAPK12 | 1556340_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5549
p=0.892
p=0.4711
p=0.6656
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPK12 | 1556340_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6516
p=0.1278
p=0.3629
p=0.3109
SmallAirway2
LargeAirway2
Trachea
Macrophage2

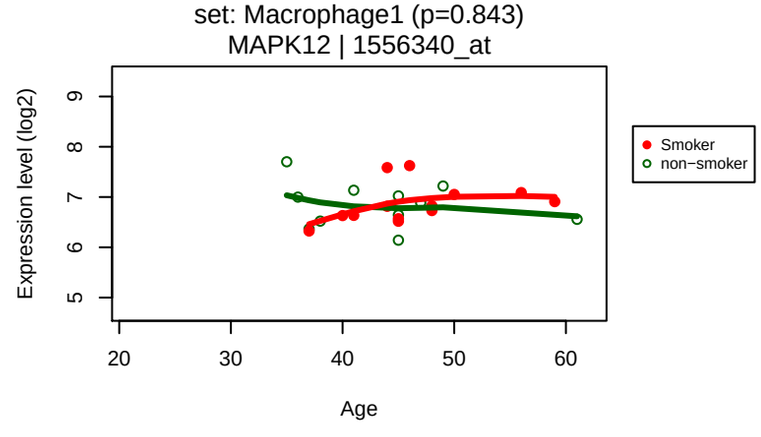

set: SmallAirway1 (p=0.027)
MAPK12 | 1556340_at
Expression level (log2)
Age
Smoker
non−smoker

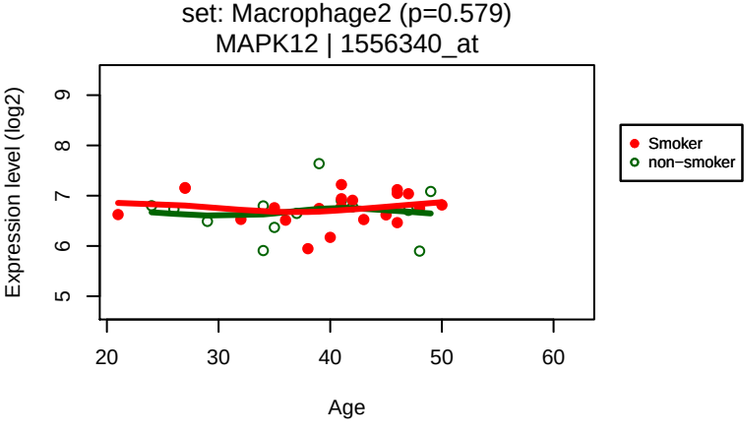

set: SmallAirway2 (p=0.4508)
MAPK12 | 1556340_at
Expression level (log2)
Age
Smoker
non−smoker

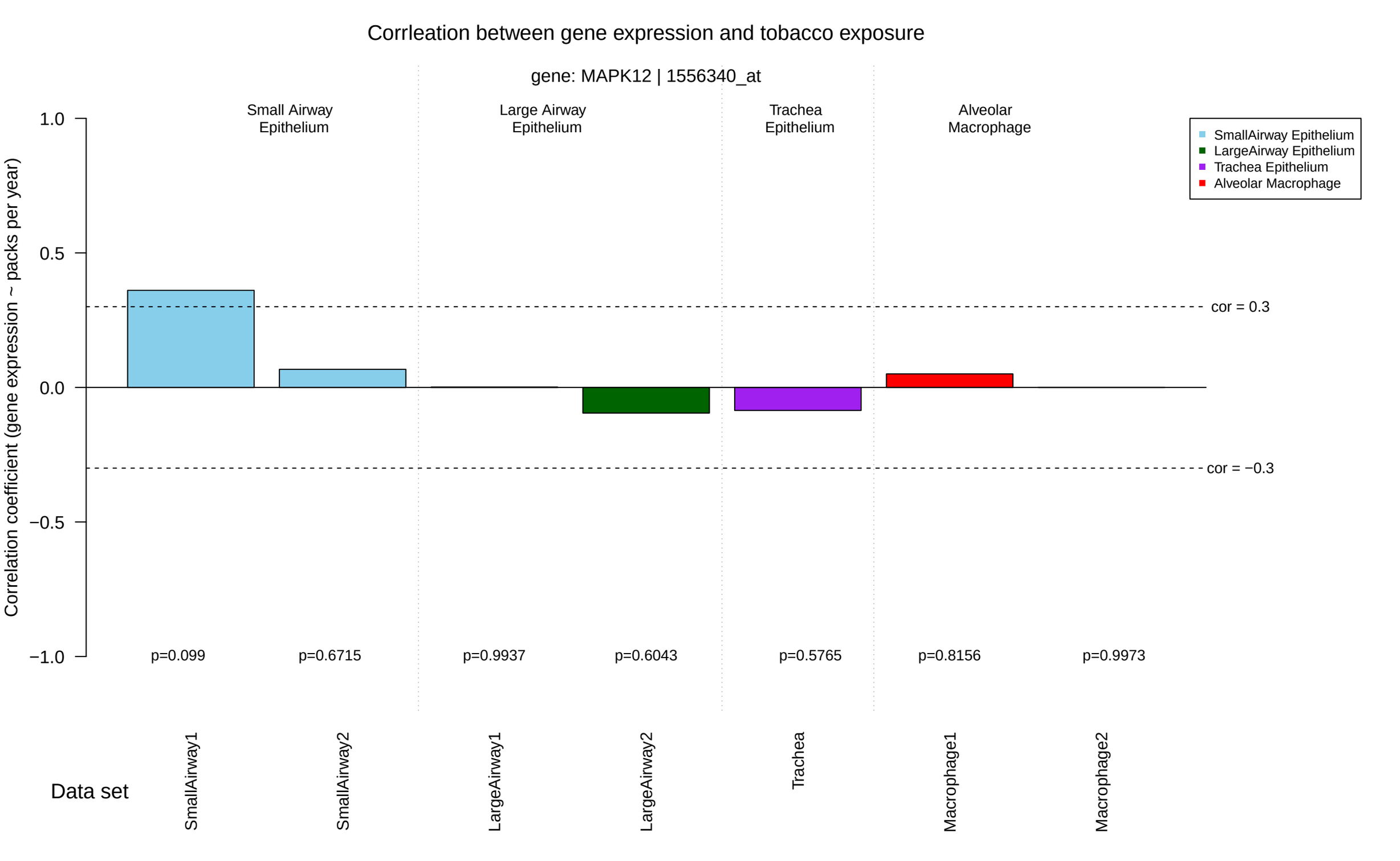

set: LargeAirway1 (p=0.2777)
MAPK12 | 1556340_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4522)
MAPK12 | 1556340_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9878)
MAPK12 | 1556340_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.843)
MAPK12 | 1556340_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.579)
MAPK12 | 1556340_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPK12 | 1556340_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.099
p=0.6715
p=0.9937
p=0.6043
p=0.5765
p=0.8156
p=0.9973
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2